ML / DL REAL – WORLD APP

GROUP C

AUTHORS: HUMPY PININTI, SHREYAS KOTIAN, BHARATH PUNATI, NGUYEN, MAX LARYEA, ATAUR REHMAN, PRASANTH MAMIDISETTI

WHAT IS MACHINE LEARNING?

- Subset of artificial intelligence
- Core Idea- Feed examples >> Identify pattern >> Predict
- Why is it better than traditional programming.
- Different types
  - Supervised- give examples
  - Unsupervised- find hidden patterns
  - Reinforcement- trial and error

- Limitations
  - Lots of data
  - Biased
  - Overfitting- Memorize data instead of learning

MACHINE LEARNING VS DEEP LEARNING

- **Machine Learning:**
- It learns from data to make predictions
- Humans tell it which information is important
- Works well with small and organized data
- Trains fast and does not need heavy computers
- Easy to understand why it made a decision
- Good for simple tasks like spam detection and price prediction

- **Deep Learning:**
- It learns patterns on its own without human help
- Figures out what information is important by itself
- Needs a very large amount of data to work well
- Takes longer to train and needs powerful computers
- Hard to understand how it made a decision
- Good for complex tasks like recognizing faces and understanding speech

PRODUCT RECOMMENDATION- ONLINE SHOPPING

ALGORITHM: K-NEAREST NEIGHBORS

- **How it works**
- An online store studies past customers behavior such as products viewed, items purchased, and similar user preferences. The model compares patterns across customers to recommend products a shopper is likely to want next

- **Main Features Used**

Browsing history, purchase history, product rating, cart activity, and similarity between users or items.

- **Business value**

Improve customer experience, increases sales, and helps businesses show more relevant products at the right time

FAKE REVIEW DETECTION

Business Impact

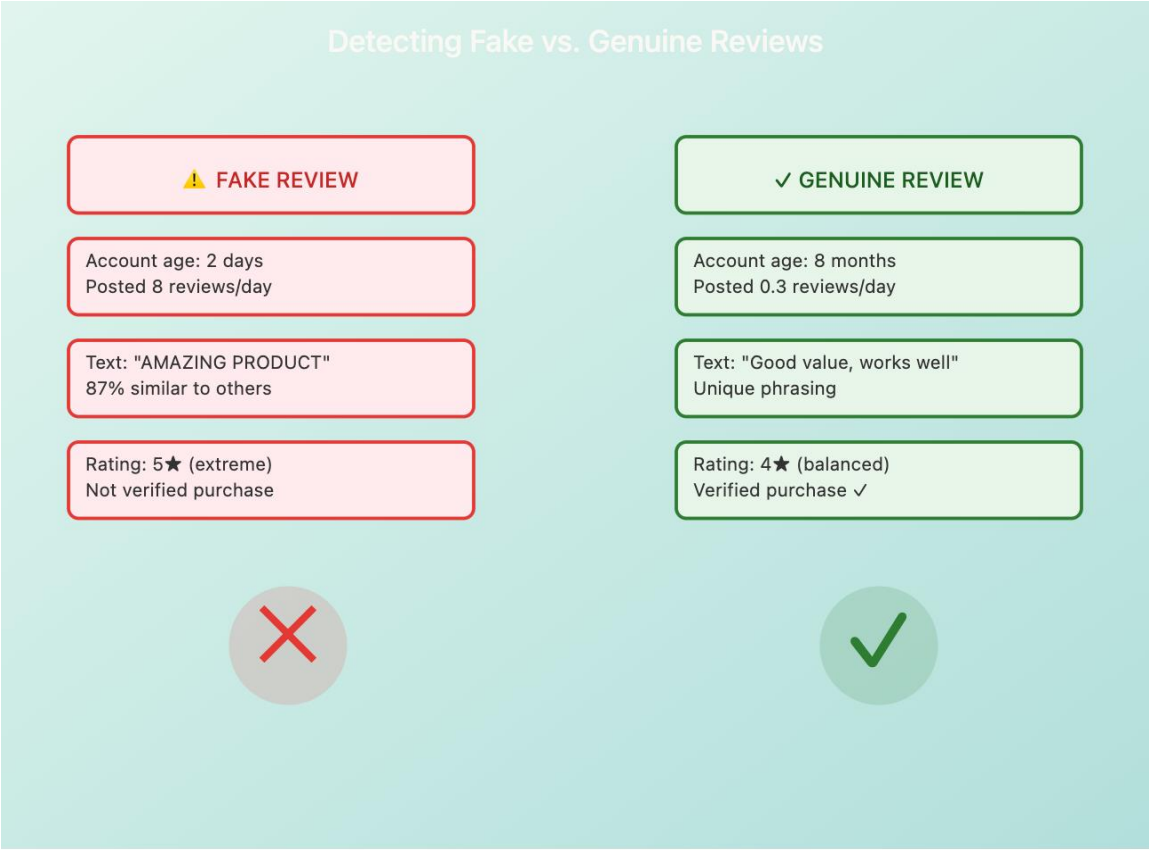
- Millions of fraudulent reviews posted daily across platforms
- Business revenue affected by skewed product ratings
- Legal liability and regulatory compliance risks

Key Features

- Review text content and metadata (date, rating, reviewer profile)
- User behavior patterns (purchase history, review frequency)
- Network analysis (reviewer connections, suspicious clustering)

Classification Model

- Supervised Learning: train on labeled and gives an binaryoutput flag
- Random Forest model



WHAT IS DEEP LEARNING?

- Subset of machine learning
- Uses multiplayer neural networks to automatically analyze data
- Handles multiple sources like images, text, speech
- Needs large datasets and computing power
- Powers modern AI systems: AI assistants, image recognition, autonomous driving

APPLICATION OF DEEP LEARNING IN HEALTHCARE RESEARCH

Problem:
Thyroid follicle thickness → key diagnostic indicator

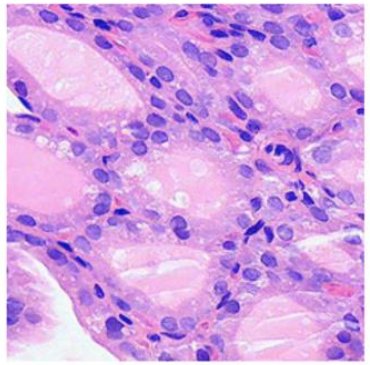
Challenge:
Manual measurement → time-consuming & inconsistent

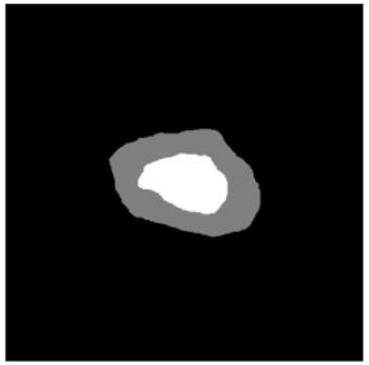
Solution: Convolutional Neural Networks

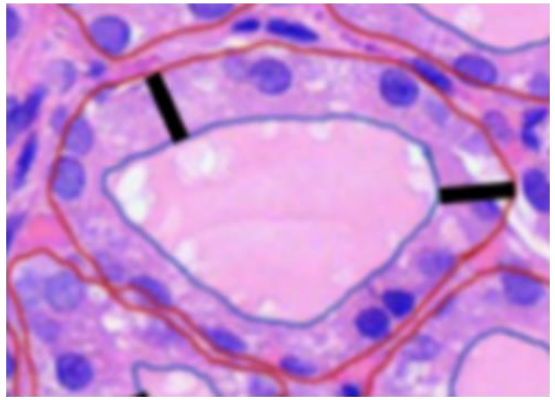
CNN: Learns visual patterns (edges → textures → structures)

Approach:
Automatic feature extraction → no manual engineering

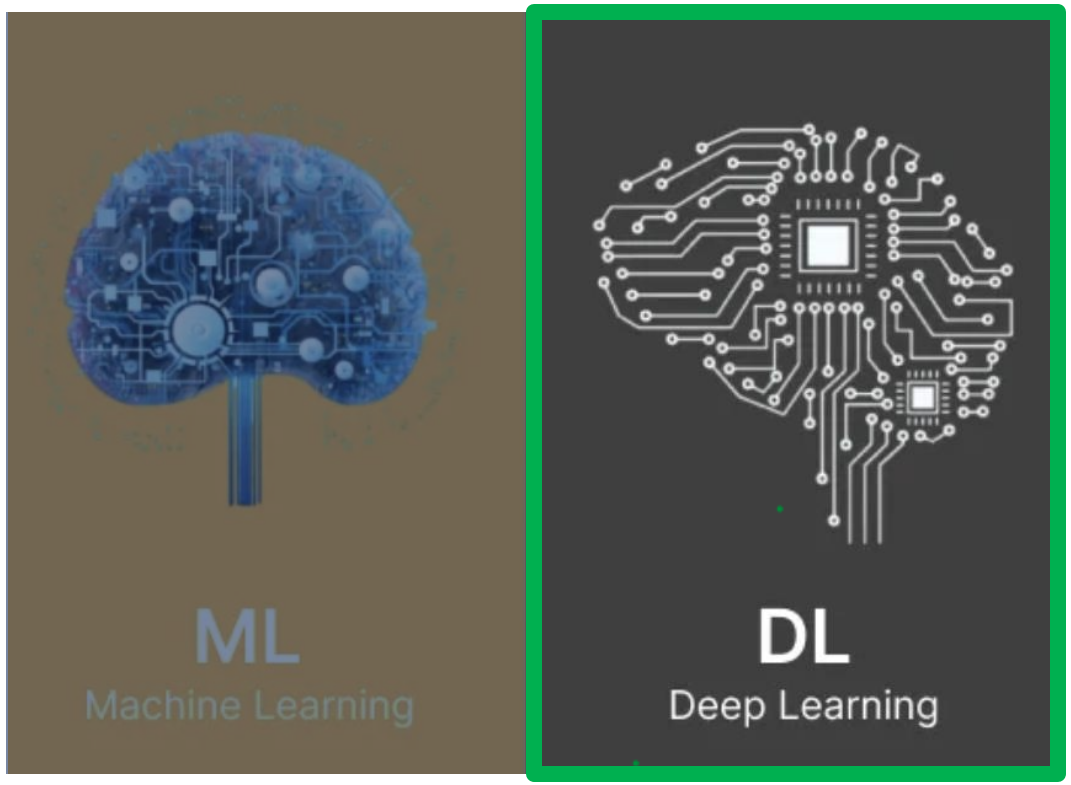
Impact:
Faster, consistent, and scalable predictions







WHY DL USE CASE ISN'T FIT FOR ML



Key Differences:

- **Data structure:** DL algos can create results from unstructured data while ML thrives on structure
- **Interpretability:** DL results tend to be black boxes, making auditing more complex than ML results

Unique Application:

- **CNN use case:** CNNs have strong use cases in healthcare, making it fit for thyroid diagnostic imaging
- **Cellular structure:** the thyroid cell changes structure frequently, making it unfit for ML applications

Summary: all DL is ML, but not all ML is DL.

THANK YOU